



Université Claude Bernard



Lyon 1

Data & Intelligence for Smart Systems

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French-Italian Language Processing tasks



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1 Introduction

Cross-lingual word embeddings enable knowledge transfer between languages by mapping words into shared semantic spaces. This capability is particularly valuable for low-resource languages and multilingual natural language processing tasks. This project explores embedding generation, alignment, and evaluation for French and Italian, two Romance languages with substantial lexical overlap but distinct morphological patterns.

As part of the Natural Language Processing course taught by Dr. Andrea Mauri (January-March 2025), I implemented and compared five embedding methods: one-hot encoding, TF-IDF, Word2Vec [5], FastText [2], and GloVe [6]. I then applied Procrustes alignment [7] using the MUSE bilingual dictionary [3] to map French and Italian embeddings into a shared space. The quality of aligned embeddings was evaluated through translation accuracy, visualization, and downstream tasks including language identification and sentiment analysis.

The choice of French and Italian was motivated by practical considerations: I begin an internship in Padua in mid-March and am actively learning Italian. Comparing these languages seemed an excellent way to fail at both simultaneously—though I hoped the embeddings would perform better than my conversational skills.

2 Embedding Generation

Word embeddings map words to dense vector representations that capture semantic relationships. I generated five types of embeddings for both French and Italian using the Tatoeba parallel corpus.

2.1 *Corpus and Preprocessing*

I used the Tatoeba French-Italian parallel corpus [9] containing 10,000 sentence pairs. Preprocessing included tokenization, lowercasing, and removal of punctuation using NLTK's word tokenizer for both languages. This resulted in vocabularies of approximately 6,450 unique tokens per language.

2.2 *Embedding Methods*

Table 1 summarizes the five embedding approaches.

Method	Dimension	Vocabulary	Source
One-hot	6,450	6,450	Tatoeba
TF-IDF	10,000	5,000	Tatoeba
Word2Vec	100	3,521 (FR) / 3,579 (IT)	Tatoeba
FastText	100	3,521 (FR) / 3,579 (IT)	Tatoeba
GloVe	300	5,847 (FR) / 6,526 (IT)	Common Crawl

Table 1: Embedding configurations. Vocabulary indicates words retained after minimum frequency filtering (min_count=2 for Word2Vec and FastText).

One-hot encoding represents each word as a binary vector of vocabulary size, with exactly one dimension set to 1. This sparse representation captures no semantic information but serves as a baseline.

TF-IDF (Term Frequency-Inverse Document Frequency) [8] weights words by their frequency in a document relative to the entire corpus. I used scikit-learn’s TfidfVectorizer with max_features=5000, resulting in 10,000-dimensional vectors (one dimension per document).

Word2Vec [5] learns dense representations by predicting context words. I trained CBOW (Continuous Bag-of-Words) models with vector_size=100, window=5, min_count=2, and 10 epochs using gensim. The small corpus resulted in vocabularies of approximately 3,500 words per language.

FastText [2] extends Word2Vec by incorporating subword information through character n-grams (min_n=3, max_n=6). I used the same training parameters as Word2Vec. This allows approximate representations for out-of-vocabulary words through their character components.

GloVe (Global Vectors) [6] learns embeddings from word co-occurrence statistics. Rather than training on Tatoeba, I used pre-trained vectors from Common Crawl [4] (840B tokens). I loaded the cc.fr.300.vec and cc.it.300.vec files and extracted only words present in the Tatoeba vocabulary, resulting in 300-dimensional vectors with coverage of 5,847 French words and 6,526 Italian words.

2.3 Discussion

The choice of embedding method involves tradeoffs between semantic quality and vocabulary coverage. Corpus-trained embeddings (Word2Vec, FastText) achieved limited vocabulary coverage (approximately 3,500 words) due to the small training corpus. Pre-trained GloVe embeddings provided broader coverage (5,800+ words) and richer semantic structure, as demonstrated in the analysis

section. FastText’s subword modeling offers an advantage for morphologically rich languages by handling out-of-vocabulary words, though this benefit was not fully realized on the small Tatoeba corpus

3 Procrustes Alignment

Cross-lingual word embeddings enable transfer learning between languages by mapping words from different languages into a shared vector space. I used Procrustes analysis [7] to align French and Italian embeddings based on a bilingual dictionary.

3.1 Methodology

Procrustes alignment finds an orthogonal transformation matrix that minimizes the distance between paired word vectors. Given a bilingual dictionary with n word pairs (w_i^{fr}, w_i^{it}) , let $X \in \mathbb{R}^{n \times d}$ and $Y \in \mathbb{R}^{n \times d}$ be matrices of French and Italian word vectors respectively. The alignment finds an orthogonal matrix W such that:

$$W^* = \arg \min_W \|XW - Y\|_F \quad \text{subject to} \quad W^T W = I$$

The solution is obtained through Singular Value Decomposition: if $Y^T X = U \Sigma V^T$, then $W^* = UV^T$.

I used the MUSE French-Italian bilingual dictionary [3] containing 5,000 translation pairs. Following the MUSE protocol, I applied iterative refinement: after initial alignment, I selected high-confidence pairs (top 75% by cosine similarity) and re-aligned for 5 iterations. Alignment quality was measured using Precision at 1 (P@1): the percentage of source words whose nearest neighbor in the target space is the correct translation. I used Cross-domain Similarity Local Scaling (CSLS) [3] to compute nearest neighbors, which mitigates the hubness problem where certain words become nearest neighbors for many queries.

3.2 Results and Discussion

Table 2 presents alignment results on a held-out test set of 1,500 word pairs.

GloVe achieved strong performance (87.4% test accuracy), demonstrating effective cross-lingual mapping. TF-IDF achieved 76.8% accuracy despite using document-level features. Word2Vec and FastText both failed (0% test accuracy)

Embedding	Train P@1	Test P@1	Avg Cosine Sim
GloVe	0.842	0.874	0.682
TF-IDF	0.752	0.768	0.877
Word2Vec	0.156	0.000	0.949
FastText	0.020	0.000	0.992

Table 2: Procrustes alignment results. P@1 measures translation accuracy using CSLS nearest neighbors.

due to the small Tatoeba corpus. With only 10,000 training sentences, these embedding spaces lack sufficient structure for reliable alignment. After 5 refinement iterations, the anchor set reduced from 1,232 pairs to 390 pairs, indicating most word pairs were discarded as unreliable. These results show that alignment quality depends critically on training data size: pre-trained embeddings from large corpora support accurate cross-lingual mapping, while embeddings from small corpora cannot establish reliable correspondences.

4 Visualization

(Almost all visualizations are one page below)

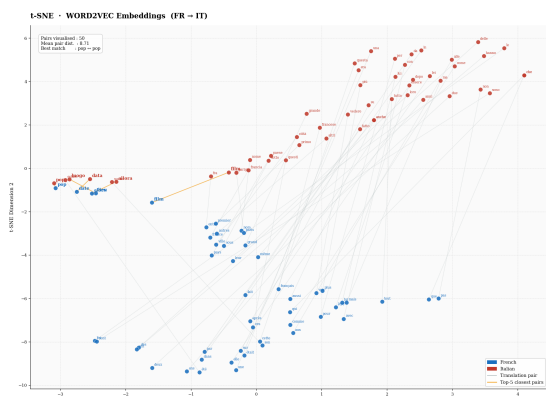
To understand embedding structure, I visualized French and Italian word spaces using t-SNE and PCA on 50 translation pairs from the MUSE dictionary.

t-SNE visualizations reveal distinct patterns. Word2Vec and FastText show strong language separation (distances 8.71 and 18.53), while GloVe shows partial overlap (5.32) despite independent training.

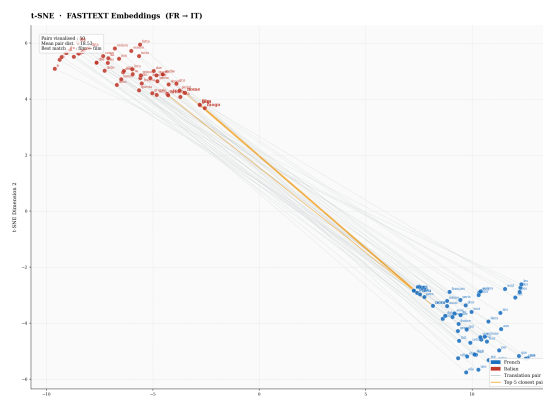
PCA reveals fundamental geometric differences. Word2Vec concentrates 77% variance in two components (PC1=47.9%, PC2=29.2%), indicating simplified structure from small corpus training. GloVe shows distributed variance (PC1=18.3%, PC2=8.6%), characteristic of large-corpus embeddings. FastText shows extreme concentration (PC1=82.2%), reflecting collapsed semantic space.

The radar chart shows GloVe’s balanced performance across five metrics. TF-IDF excels at vocabulary coverage (no frequency filtering), while Word2Vec and FastText show lower coverage (min_count=2 on small corpus).

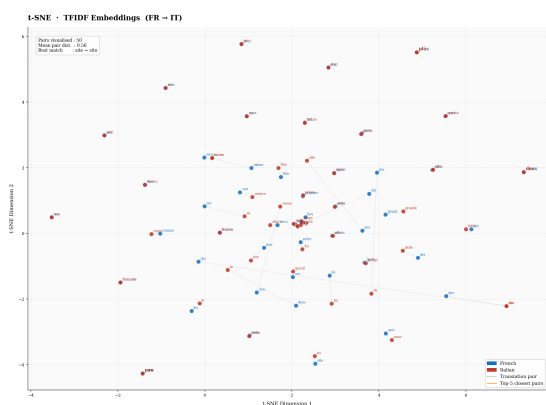
The alignment comparison reveals that high cosine similarity does not guarantee translation quality. FastText achieves 0.99 similarity but 0% translation accuracy, while GloVe achieves 0.68 similarity with 87.4% accuracy. This confirms that corpus-trained embeddings produce nearly colinear vectors that align trivially without capturing semantic correspondences.



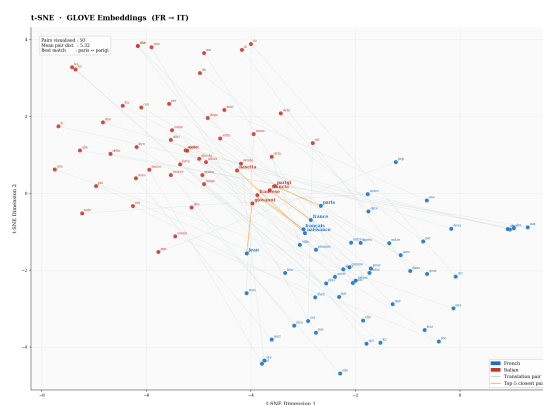
Word2Vec t-SNE (dist=8.71)



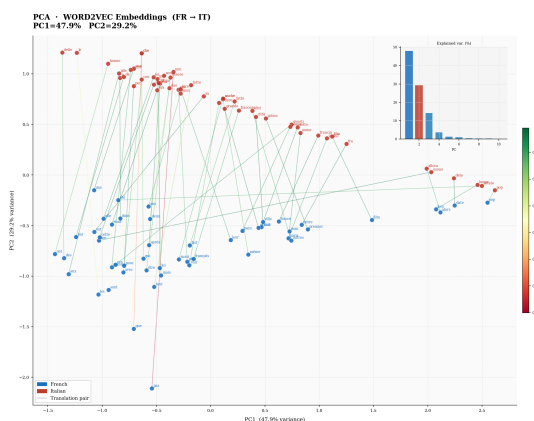
FastText t-SNE (dist=18.53)



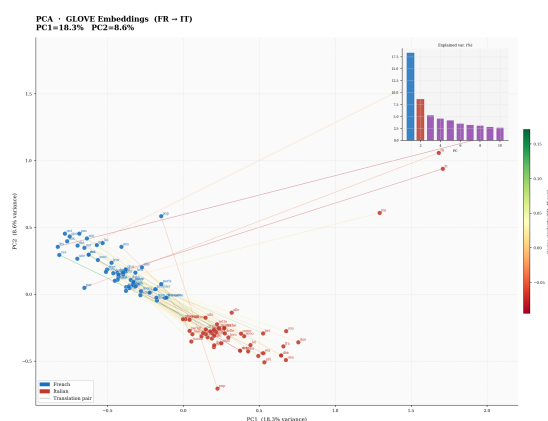
TF-IDF t-SNE (dist=0.56)



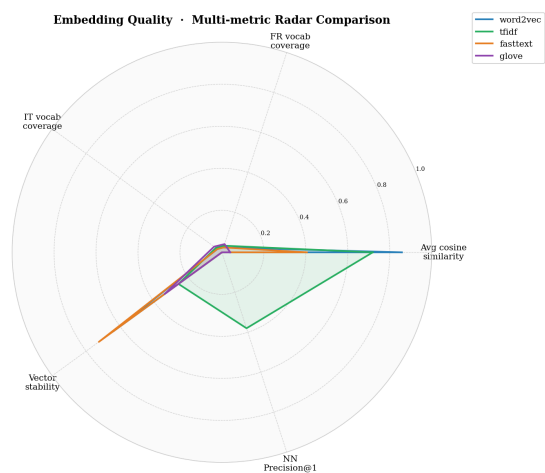
GloVe t-SNE (dist=5.32)



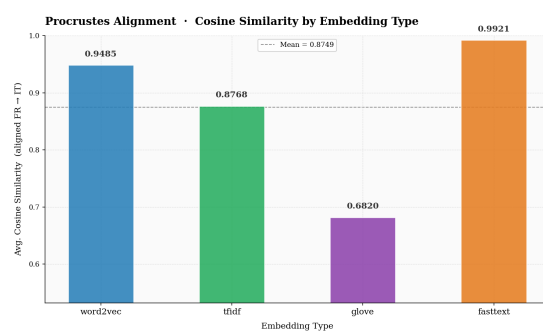
Word2Vec PCA (PC1=47.9%)



GloVe PCA (PC1=18.3%)



Multi-metric radar chart



Procrustes alignment quality

5 Embedding Quality Analysis

To assess embedding quality, I analyzed synonym detection, cross-lingual cognate similarity, polysemy representation, and out-of-vocabulary handling.

5.1 Synonym and Antonym Detection

Table 3 shows nearest neighbors for selected French words.

Embedding	Query	Top-3 Neighbors (similarity)
Word2Vec	bien	ça (0.995), parler (0.995), dire (0.994)
Word2Vec	heureux	sans (0.998), non (0.998), regarde (0.998)
GloVe	bien	aussi (0.786), vraiment (0.754), mal (0.753)
GloVe	heureux	heureuse (0.713), malheureux (0.682), ravi (0.680)
GloVe	grand	petit (0.720), beau (0.627), seul (0.626)

Table 3: Nearest neighbors for French words. GloVe captures semantic relationships (synonyms and antonyms), while Word2Vec returns high-frequency words.

Word2Vec shows very high similarities (0.99-1.0) between neighbors with limited semantic coherence. GloVe captures meaningful relationships, retrieving "malheureux" (unhappy) as an antonym and "ravi" (delighted) as a synonym for "heureux" (happy).

5.2 Cross-lingual Cognate Similarity

Table 4 presents cognate similarities before alignment.

French	Italian	Word2Vec	FastText	GloVe
porte	porta	0.898	0.484	0.215
musique	musica	0.892	0.453	0.174
difficile	difficile	0.895	0.461	0.195
temple	tempio	0.893	0.451	0.182
Average		0.837	0.407	0.040

Table 4: Cosine similarity between cognates before Procrustes alignment. Word2Vec shows high similarity from parallel training data.

Word2Vec shows high cognate similarity (0.84) because both languages are trained on parallel sentences. GloVe shows low similarity (0.04) from independent corpora.

5.3 Out-of-Vocabulary Analysis

Table 5 summarizes vocabulary coverage.

Embedding	Vocab (FR/IT)	OOV Rate (FR/IT)	OOV Strategy
Word2Vec	3521 / 3579	45.4% / 48.5%	None
FastText	3521 / 3579	45.4% / 48.5%	Subword n-grams
GloVe	5847 / 6526	9.4% / 6.3%	None

Table 5: Vocabulary coverage on Tatoeba corpus. FastText handles OOV words through character n-grams.

Word2Vec and FastText cover 3,500 words each (45-48% OOV), while GloVe covers 5,800+ words (6-9% OOV). FastText approximates unseen words through character n-grams, while Word2Vec and GloVe skip them.

5.4 Polysemy: zero

Polysemy analysis showed near-zero variance in Word2Vec and FastText neighbor distributions, indicating failure to represent multiple word senses, while GloVe showed small variance suggesting some differentiation.

5.5 Discussion

Training corpus size affects embedding quality. The small Tatoeba corpus produces embeddings with limited semantic differentiation, while pre-trained GloVe better captures semantic relationships. The high cognate similarity in Word2Vec reflects parallel training rather than semantic alignment.

6 Language Identification

To evaluate the discriminative power of embeddings for language classification, I trained logistic regression classifiers to distinguish French from Italian sentences using averaged word vectors as features.

6.1 Experimental Setup

I used 10,000 French sentences and 10,000 Italian sentences from the Tatoeba corpus, representing each sentence as the mean of its word vectors. The classifier was trained on 80% of the data and tested on the remaining 20%.

6.2 Results

Table 6 presents the classification performance.

Embedding	Accuracy	F1 (FR)	F1 (IT)
Word2Vec	0.99	0.99	0.99
FastText	1.00	1.00	1.00
GloVe	1.00	1.00	1.00

Table 6: Language identification performance across embedding types. All methods achieve near-perfect accuracy.

All three embedding types achieved near-perfect performance (99-100% accuracy). FastText and GloVe reached perfect classification, while Word2Vec made only 26 errors on 3,991 test samples. This task is relatively easy because French and Italian, despite being Romance languages, have distinct lexical distributions that are readily captured even by simple averaged word vectors.

7 Cross-lingual Sentiment Analysis

To evaluate the practical utility of multilingual embeddings and alignment methods, I conducted sentiment analysis experiments on French and Italian review data. The goal was to assess whether a classifier trained on French sentiment data could predict sentiment in Italian texts, and whether Procrustes alignment would improve cross-lingual transfer.

7.1 Dataset and Experimental Setup

I used the Allocine movie review dataset [1] for French (5,000 samples: 2,500 positive, 2,500 negative) and Italian Twitter data (2,022 samples: 1,011 positive, 1,011 negative). Each review was represented as the mean of its word vectors. I trained logistic regression classifiers in three settings: monolingual French, monolingual Italian, and cross-lingual (train on French, test on Italian), comparing raw versus Procrustes-aligned embeddings.

7.2 Results

Table 7 presents classification accuracy across different conditions.

Embedding	Experiment	Accuracy	Macro F1
GloVe	Monolingual FR	0.73	0.73
GloVe	Monolingual IT	0.69	0.69
GloVe	Cross-lingual (raw)	0.60	0.58
GloVe	Cross-lingual (aligned)	0.58	0.58
Word2Vec	Monolingual FR	0.63	0.63
Word2Vec	Monolingual IT	0.58	0.58
Word2Vec	Cross-lingual (raw)	0.49	0.34
Word2Vec	Cross-lingual (aligned)	0.49	0.35

Table 7: Sentiment classification results. Cross-lingual experiments involve training on French data and testing on Italian data.

GloVe achieved 70% monolingual accuracy and 60% cross-lingual performance without alignment, notably above the 50% random baseline. However, Procrustes alignment decreased accuracy to 58%. Word2Vec showed lower monolingual performance (60%) and cross-lingual transfer near random (49%), with marginal F1 improvement from alignment (0.34 to 0.35).

Table 8 reveals the per-class performance breakdown.

Embedding	Experiment	Negative		Positive
		Precision	Recall	Precision
GloVe	Cross (raw)	0.57	0.78	0.65
GloVe	Cross (aligned)	0.57	0.65	0.59
Word2Vec	Cross (raw)	0.22	0.01	0.49
Word2Vec	Cross (aligned)	0.39	0.02	0.50

Table 8: Per-class performance for cross-lingual transfer. Procrustes alignment improves Word2Vec negative precision from 0.22 to 0.39.

Word2Vec exhibits strong bias toward positive predictions (recall 0.01 on negatives). Procrustes alignment improves negative precision from 0.22 to 0.39, suggesting better cross-lingual correspondences for sentiment vocabulary.

7.3 Discussion

These results show that Procrustes alignment utility depends on embedding properties. Pre-trained GloVe embeddings already exhibit cross-lingual correspondences, and forced alignment may disrupt existing structures. In contrast, Word2Vec trained on smaller corpora benefits from explicit alignment, particularly in correcting prediction biases. The MUSE dictionary may lack sentiment-bearing terms, limiting cross-lingual transfer. A sentiment-specific alignment dictionary would likely improve performance.

8 Conclusion

This project explored multilingual word embeddings for French and Italian across five representation methods. Pre-trained GloVe embeddings demonstrated superior semantic quality compared to embeddings trained on the small Tatoeba corpus. Procrustes alignment achieved strong performance on GloVe ($P@1=0.87$) but failed on corpus-trained embeddings due to insufficient training data. Cross-lingual sentiment transfer experiments revealed that alignment utility depends on initial embedding properties: pre-trained embeddings already exhibit natural correspondences, while small-corpus embeddings benefit from explicit alignment. The use of general translation dictionaries for sentiment-specific tasks remains a limitation.

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